

ASP.NET Core MVC - Beautifying with Bootstrap

Bootstrap 2 (Beautifying ASP.NET Application with Bootstrap)

Overview

Bootstrap is a front-end framework used to design and beautify web applications by providing ready-made CSS styles, layouts, and responsive components.

In **ASP.NET Core applications**, Bootstrap helps developers create:

- Clean user interfaces
- Responsive layouts
- Styled forms and tables
- Modern-looking web pages
- Mobile-friendly applications

Simple Idea

ASP.NET Core handles the **backend logic**, while Bootstrap handles the **appearance and layout** of the application.

Role of HTML and CSS in ASP.NET Core Application

HTML in ASP.NET Core

HTML (HyperText Markup Language) is used to structure the web page.

It defines:

- Headings
- Paragraphs
- Forms
- Tables
- Buttons
- Links
- Layout

Example

```
<h1>Student Information</h1>

<form>
  <input type="text" />
  <button>Submit</button>
</form>
```

Role of HTML

- Creates the structure of the page
- Defines elements
- Displays content

- Works inside **Razor Views (.cshtml)**
-

CSS in ASP.NET Core

CSS (Cascading Style Sheets) is used to style HTML elements.

It controls:

- Colors
- Fonts
- Layout
- Spacing
- Design
- Responsiveness

Example

```
h1 {  
  color: blue;  
}  
  
button {  
  background-color: green;  
  color: white;  
}
```

HTML + CSS in ASP.NET Core

In ASP.NET Core:

Component	Role
HTML	Structure
CSS	Design
Bootstrap	Ready-made CSS framework
ASP.NET Core	Backend logic

Razor View Example

```
<h2 class="text-primary">Student List</h2>  
  
<button class="btn btn-success">Add Student</button>
```

Here:

- HTML creates the button
 - Bootstrap styles the button
 - ASP.NET Core processes the logic
-

Characteristics of Front-End Frameworks

Definition

A **front-end framework** is a collection of pre-written HTML, CSS, and JavaScript used to design user interfaces quickly and efficiently.

Characteristics

1. Responsive Design

Automatically adjusts to:

- Desktop
- Tablet
- Mobile

Example:

A page looks good on both laptop and phone.

2. Predefined CSS Classes

Developers use ready-made classes.

Example:

```
btn btn-primary
```

Instead of writing CSS manually.

3. Grid System

Helps organize layout into rows and columns.

Example:

```
row and col-md-6
```

4. Reusable Components

Includes:

- Buttons
 - Forms
 - Tables
 - Navigation bars
 - Cards
 - Alerts
-

5. Faster Development

No need to design from scratch.

Just use framework classes.

Different Front-End Frameworks Available

1. Bootstrap

Most popular framework.

Features:

- Responsive design
- Easy to use
- Large community
- Many components

Used in ASP.NET Core projects.

2. Tailwind CSS

Utility-based framework.

Features:

- Flexible design
- Custom styling
- Lightweight

Example:

```
bg-blue-500 text-white p-4
```

3. Foundation

Advanced framework.

Features:

- Professional layout
 - Flexible components
 - Responsive grid
-

4. Bulma

Modern CSS framework.

Features:

- Simple
 - Lightweight
 - Mobile-first
-

Comparison

Framework	Difficulty	Flexibility	Popularity
Bootstrap	Easy	Medium	High
Tailwind	Medium	High	High
Foundation	Hard	High	Medium
Bulma	Easy	Medium	Medium

Bootstrap and Its Grid System

What is Bootstrap?

Bootstrap is a **CSS framework** used to design responsive web pages.

It provides:

- Layout system
 - Components
 - Utilities
 - JavaScript plugins
-

Bootstrap Grid System

Definition

The **Bootstrap Grid System** divides the screen into **12 columns**.

This helps organize layout.

Structure

container

row

column

Example

```
<div class="container">
  <div class="row">
    <div class="col-md-6">Left</div>
    <div class="col-md-6">Right</div>
  </div>
</div>
```

Output

Screen divided into two equal parts.

Left | Right

12 Column Rule

Columns	Size
col-md-12	full width
col-md-6	half
col-md-4	1/3
col-md-3	1/4

Responsive Sizes

Class	Device
col-sm	Small
col-md	Medium
col-lg	Large
col-xl	Extra Large

Bootstrap Features

1. Responsive Design

Works on all screen sizes.

2. Grid System

Organizes layout.

3. Components

Includes:

- Navbar
 - Buttons
 - Forms
 - Tables
 - Cards
 - Alerts
-

4. Utility Classes

Example:

text-center
bg-primary
mt-3
p-2

Used for spacing and styling.

5. Mobile First

Designed for mobile devices first.

CSS Classes for Form Elements

Bootstrap provides ready-made form styling.

Input Elements

Example

```
<input type="text" class="form-control" />
```

Output

Styled textbox.

Common Classes

Class	Purpose
form-control	textbox styling
form-label	label styling
form-group	group form elements
form-check	checkbox/radio
form-select	dropdown
btn	button
btn-primary	blue button

Example Form

```
<div class="mb-3">
<label class="form-label">Name</label>
<input type="text" class="form-control">
</div>

<div class="mb-3">
<label class="form-label">Course</label>
<select class="form-select">
<option>BSIT</option>
<option>BSCS</option>
</select>
</div>

<button class="btn btn-primary">Submit</button>
```

Select Elements

Example

```
<select class="form-select">
<option>Male</option>
<option>Female</option>
</select>
```

Output

Styled dropdown.

CSS Classes for Tables

Bootstrap also beautifies tables.

Basic Table

```
<table class="table">
```

Table Classes

Class	Purpose
table	basic table
table-striped	striped rows
table-bordered	borders
table-hover	hover effect
table-dark	dark table
table-primary	colored table

Example


```
<table class="table table-striped table-bordered">
<thead>
<tr>
<th>ID</th>
<th>Name</th>
<th>Course</th>
</tr>
</thead>
<tbody>
<tr>
<td>1</td>
<td>John</td>
<td>BSIT</td>
</tr>
</tbody>
</table>
```

Using Bootstrap in ASP.NET Core Application

Step 1: Install Bootstrap

Using CDN

Add inside **_Layout.cshtml**

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
```

Step 2: Add Bootstrap Script

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
```

Step 3: Use Bootstrap in Layout

Open:

```
Views/Shared/_Layout.cshtml
```

Add Bootstrap.

Example Layout

```
<!DOCTYPE html>
<html>
<head>
<title>Student App</title>

<link href="bootstrap.css" rel="stylesheet" />
</head>

<body>

<div class="container">
  @RenderBody()
</div>

</body>
</html>
```

Example ASP.NET Core Page

```
<h2 class="text-primary">Student Form</h2>

<div class="mb-3">
  <label>Name</label>
  <input class="form-control">
</div>

<button class="btn btn-success">Save</button>
```

Result

Styled ASP.NET Core page with:

- Blue heading
- Styled textbox
- Green button
- Clean layout

Real-World Example

Without Bootstrap:

Plain and unattractive page.

With Bootstrap:

Professional-looking web application.

Like:

Without Bootstrap	With Bootstrap
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Plain form	Styled form
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Basic table	Professional table
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Simple layout	Responsive layout
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Summary

HTML and CSS

- HTML builds structure
- CSS styles design
- Bootstrap enhances appearance

Front-End Frameworks

- Bootstrap
- Tailwind
- Foundation

- Bulma

They help create responsive and modern UI.

Bootstrap

- Grid system
 - Components
 - Responsive design
 - Utility classes
-

Form Classes

- form-control
 - form-select
 - btn
 - form-label
-

Table Classes

- table
 - table-striped
 - table-bordered
 - table-hover
-

Using Bootstrap

1. Add Bootstrap CDN
2. Add to Layout
3. Use classes in Razor views
4. Build responsive UI